Pierson Jones

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EDUCATION

Duke University | B.E. Electrical and Computer Engineering & B.S. Computer Science **Aug 2025 – May 2029**
GPA: 3.962 | Dean's List with Distinction (Spring 2026)
Relevant Coursework: Data Structures and Algorithms, Signals and Systems, Fundamentals of Electrical and Computer Engineering, Computing for Engineers, Engineering Design and Communication

SKILLS

Programming: Python, Java, C++, Git, MATLAB
Embedded / Firmware: AVR hardware timers (CTC mode, ISRs), UART / software-serial communication, Arduino, ESP32, XBee
Hardware: Circuit prototyping, sensor integration, multiplexers, 3D Time-of-Flight, Hall effect, QTI
Tools & Simulation: Tinkercad Circuits, Wokwi, oscilloscope, benchtop multimeter, soldering
CAD & Manufacturing: Autodesk Fusion 360, SolidWorks, Onshape, KiCad, CNC Centroid Mill, CNC Tool-pathing, Prusa

PROJECTS

Scissor Lift Collision Detection System | *Project Lead* | *Industry Partner: Skanska* | Github-ScissorLift **Aug 2025 – Present**
- Leading a team of 4 designing a microcontroller-based object detection system to reduce scissor lift collisions.
- Programmed Arduino and ESP microcontrollers in C++ for real-time sensor data acquisition and detection.
- Integrated six 3D Time-of-Flight sensors utilizing I2C communication with a multiplexer.
- Implemented threshold-based alert logic for obstacle detection and conducted iterative system testing.
- Collaborated with a power consultant and Skanska to optimize power consumption and refine system constraints for real-world deployment.
- Reduced product energy consumption from 627 mAh to 109.2 mAh over a 10-hour runtime, achieving an 83% efficiency gain through interfacing with the VL53L8CX sensor's APIs.
- Designed a non-blocking C++ state machine for adaptive sensor polling (1 Hz idle, 6 Hz active) with hysteresis-based mode switching and smoothed buzzer feedback transitions.

Azimuth Position Encoder Replication Project | *Independent Contract Project, Collins Aerospace (RTX)* **Aug 2026 – Present**
- Reverse-engineered the timing behavior of a legacy analog encoder circuit and derived the register values needed to reproduce it digitally, correcting a data-entry error in the client-supplied specification along the way.
- Programmed an Arduino hardware timer (Timer1) in CTC mode with a compare-match interrupt to generate precise signal timing, replacing an analog RC oscillator with register-driven digital logic.
- Diagnosed a simulator limitation across two test platforms (Tinkercad and Wokwi) and redesigned the measurement method around a single-board self-loopback interrupt.
- Achieved under 2% timing error against target, versus 8-13% for the original analog circuit.

Duke Robotics Team | *RoboSub Competition* **Aug 2025 – Present**
- Contributing to the design of two AUVs competing at the annual RoboSub competition.
- Designing an independent claw/gripper concept for the team's third robot using Autodesk Fusion 360 and SolidWorks.
- Leading and onboarding a team of 3 new members on the claw/gripper subteam, teaching CAD fundamentals and guiding each member through designing their own claw/gripper concept for a comparative build-and-test process to pick this year's best design.
- Collaborating with ECE and CS subteams to improve component integration (claw/gripper).
- CNC machined and fabricated buoyancy components via CNC tool-pathing in Autodesk Fusion.

ISS Deorbit Simulation | *Team of 4* | Github-ISS-Deorbit **Aug 2025 – Dec 2025**
- Worked in a team of 4 to build an orbital dynamics simulation modeling ISS deorbit, reentry, and rocket ascent.
- Implemented 4th-order Runge-Kutta integration to solve position-velocity differential equations.
- Modeled gravitational and atmospheric drag forces and implemented event detection with interpolation for stage separation and impact.
- Visualized trajectories and altitude profiles using NumPy and Matplotlib.

WORK EXPERIENCE

Rogers Park Golf Course | *Outside Operations & Guest Services Associate* **Sep 2023 – Present**
- Maintained equipment and facilities in a high-traffic environment.
- Provided customer service and ensured operational safety.

LEADERSHIP

Keys to Kindness Kind Camp | *Camp Counselor (Volunteer)* **June 2019 – Present**
- Worked with campers in 5th grade and below, promoting kindness, character, and personal growth.
- Organized and communicated with volunteers in 6th grade and older to maintain smooth operation.